



JAIPUR NATIONAL UNIVERSITY

Semester/Year End (Main/Back) Examinations, May-2025

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Computer Science Engineering) / B.Tech. CSE (Hons.) in Cyber Security / B.Tech. CSE (Hons.) in Artificial Intelligence and Machine Learning
Semester/Year:	Semester-VI
Paper Code:	BTECSCO604T22
Title of Paper:	Advanced Python
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section-A and Section-B. Section-A consists of seven questions and Section-B consists of six questions. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question Number	SECTION-A	Marks Allotted
Q.1	What are Boolean expressions in Python? Provide two examples showing their use in selection structures.	4
Q.2✓	Define a tuple in Python. How is it different from a list?	4
Q.3✓	What is the purpose of a constructor in Python classes? Write a small example.	4
Q.4✓	Explain the concept of inheritance in Python with a suitable code snippet.	4
Q.5✓	What is the role of NumPy in Python programming? Show how to create a 1-D and a 2-D array.	4
Q.6✓	What is a DataFrame in Pandas? How is it different from a Series?	4

- Q.7** Mention the use of matplotlib's scatter plot and bar chart. **4**
Where are they commonly used?

SECTION-B

- Q.1** Write a Python program that demonstrates the use of **10**
functions, iteration, and lists. Explain how control structures
help manage the program flow.
- Q.2** Explain Object-Oriented Programming concepts in Python. **10**
Demonstrate the use of class, object, method overloading,
and polymorphism with suitable examples.
- Q.3** What are generators and decorators in Python? Write **10**
examples to show how and why they are used.
- Q.4** Explain how matrix manipulation is performed using **10**
NumPy. Write a program that performs addition and
multiplication of two matrices.
- Q.5** Write a Python program to load a dataset into a Pandas **10**
DataFrame. Demonstrate column selection, row selection,
adding, and deleting data.
- Q.6** Create a program using matplotlib that displays a histogram **10**
and a box plot using sample data. Explain when each of
these plots is used.